CHAPTER 38
WATER AND WASTE

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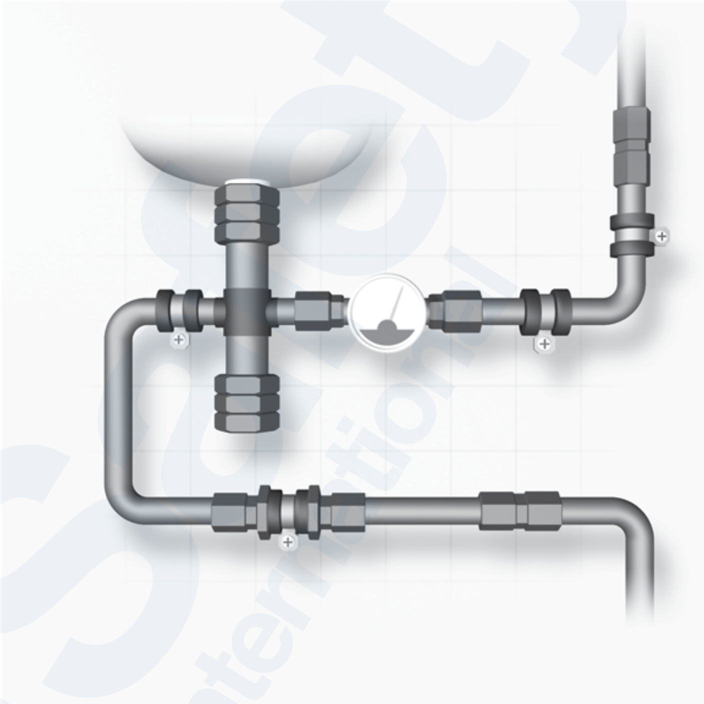
CHAPTER 38
WATER AND WASTE



38-00-00 INTRODUCTION

The water and waste system is comprised of the following sub-systems:

- Potable water
- Wash water and
- Waste water.

The three sub-systems are independent of each other so no possibility of cross contamination.

The potable water system provides cold and heated water to the optional G1 galley at the aft cabin area for either washing or drinking purposes.

The wash water system provides cold and warm water to the washroom sink located in the forward cabin area for hand washing purposes.

The waste water system supplies the water for the passenger washroom facilities on board the aircraft. The system also stores the waste water after passenger use and allows for fluid recirculation with filtration.

All three sub-systems have their own ground service panels that are located on the lower fuselage.

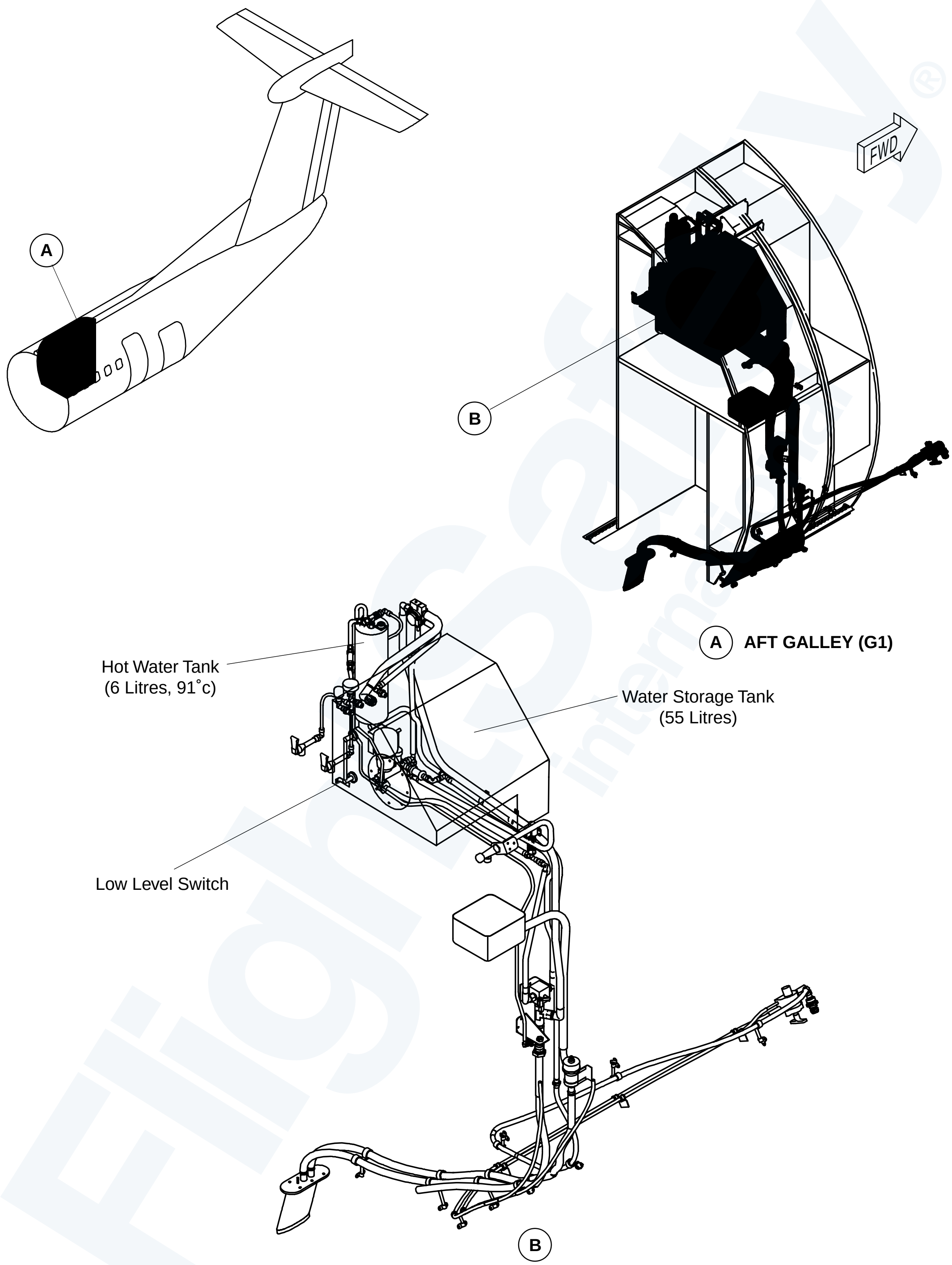


Figure 38-1. Potable Water System - Locator

38-10-00 POTABLE WATER SYSTEM

GENERAL

The potable water system supplies water to the galley. The water is used for drinking or washing purposes.

SYSTEM DESCRIPTION

Refer to Figure 38-1. Potable Water System - Locator.

The potable water system is an optional system which can be installed in aircraft with the G1 galley configuration.

It supplies heated water to the galley for in-flight catering, and unheated water for flight attendant, or sanitary purposes.

All components are installed above the floor inside the galley enclosure. The system is serviced from an external panel.

The system uses a 55 liter water storage tank, and a 6 liter hot water (boiler) tank. An electric motor drives the gear pump which supplies system pressure.

The system has:

- Two hot water taps
- One cold water tap
- One sink.

The temperature of the hot water is approximately 91°C. All water lines and components are heated by either electric heating elements or heater cuffs.

There is an optional overnight freeze protection system. External AC power operates heaters installed around the main components and water lines. This allows the water to remain in the aircraft overnight during freezing conditions.

NOTES

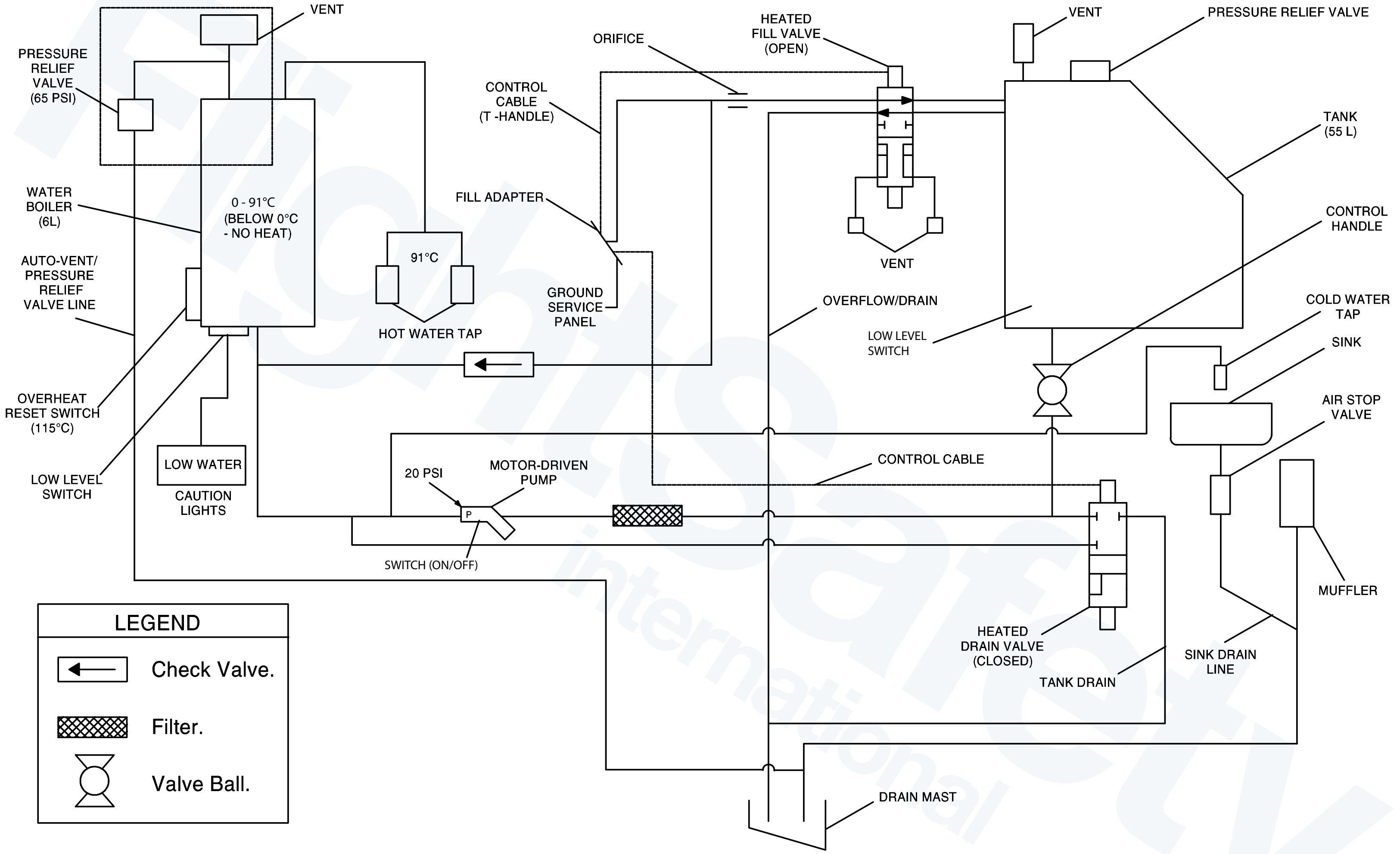


Figure 38-2. Potable Water System Schematic

Refer to Figure 38-2. Potable Water System Schematic.

A ground service cart is used to fill both tanks at the same time. The supply line from the cart is attached to the fill adapter on the external service panel.

The T-handle on the panel is turned 90 degrees in a clockwise direction. The T-handle is connected to the valve control unit located just behind the service panel. The control unit changes the turning movement of the T-handle to the linear movement of a cable which opens the filler valve.

Once the valve is opened, water flows into the fill line for the tank and boiler. A flow restrictor in the fill line to the tank makes sure there is no increase in pressure in the tank or boiler during the filling process.

The tank and boiler both have pressure relief valves to prevent overpressure. A check valve in the fill line to the boiler prevents any reverse flow through the fill adapter at the ground service panel. Water pressure during filling opens the check valve which allows water to flow into the tank and boiler.

When the hand wash faucet or a hot water tap is opened, the pressure switch in the pump senses the drop in pressure. The pump then begins transferring water from the tank into the boiler. Pumping will continue until the tap is closed by the user. A low pressure switch in the pump senses low pressure and turns the pump off to prevent it from running dry. A filter installed upstream of the pump prevents foreign objects from entering the pump and contaminating the system.

Waste water from the sink is dumped overboard through the drain mast installed in the underside of the aircraft. An airstop valve in the sink drain line is opened by the force of the waste water. The valve is normally closed to prevent cabin pressure from venting through the drain during flight.

On the ground, pulling the T-handle and rotating it 90 degree counter clockwise drains the system. The drain valve opens allowing all water in the system to dump overboard through the drain mast. Once the system is drained, the handle is centered and pushed in. A ball valve is located in the line downstream of the tank, allows the lines to be drained for maintenance purposes without having to drain the tank.

Heater cuffs protect the fill and drain valves, the pressure relief valve, the filter, the pump, the check valve, and the tank from freezing during flight. The cuffs are either molded directly to these components or are fastened by bolts. The drain mast and water lines have built in heaters. All the heaters have thermostats which control the temperature in the individual components. The heater system is energized as soon as AC power is connected to the aircraft. This prepares the system for filling by preventing flash freezing of cold water coming into contact with cold components.

An optional freeze protection system runs using external AC power. This optional system has an extra relay. When external AC power is connected, the relay is energized using DC power from either the battery bus or from a ground power unit. Once the relay is energized, the aircraft AC bus is bypassed and only the lavatory and galley heaters are energized.

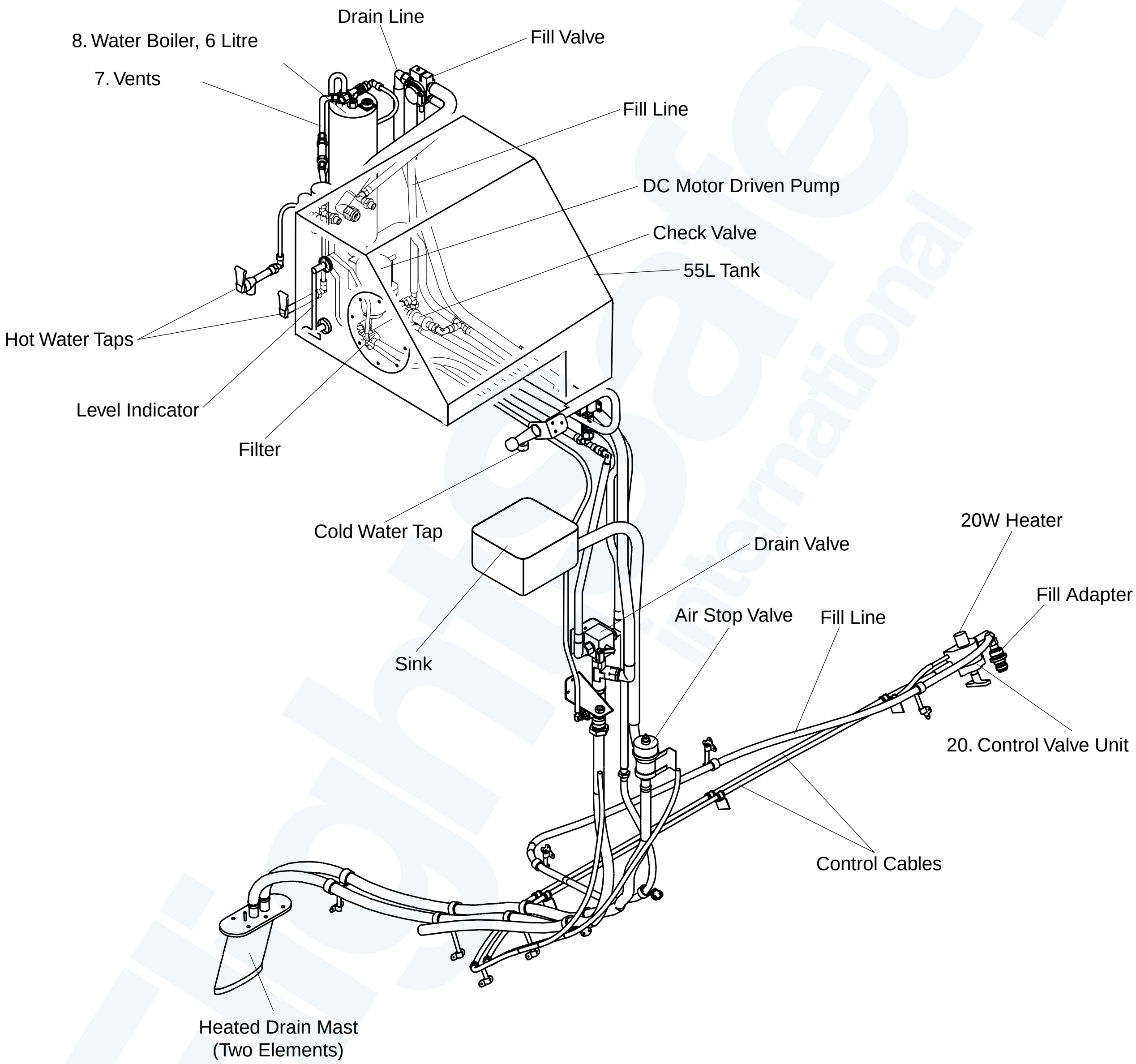


Figure 38-3. Potable Water System Detail

COMPONENTS DESCRIPTION

Galley Tank

The 55 liter welded stainless steel tank, installed in the galley, stores potable water.

The tank is filled through the fill port in the external service panel. An overflow line is located just below the fill port in the tank. The arrangement leaves 10% of the capacity unused to prevent structural damage if ice forms in the tank.

A vent and pressure relief valve on top of the tank protects the tank from overpressuring during refilling. A tank drain port is located at the low point of the tank to make sure there is no residual water in the tank after draining.

There is a sight glass on the front of the tank for a visual indication of the water level and an access cover which allows for hand cleaning of the tank.

Galley Fill Valve

Refer to Figure 38-3. Potable Water System Detail.

The fill valve allows water to flow (if greater than 15 psig) from the service panel fill line to the tank. It also allows overflow water from the tank to flow through to the drain mast.

The heated fill valve is a 4-way, two-position valve. The valve is mechanically operated from the rotary control box located at the ground service panel.

In the open position, the valve allows water to flow from the fill line to the tank. It also allows overflow water from the tank to flow through to the drain mast.

In the closed position, the valve isolates the tank from outside ambient pressure.

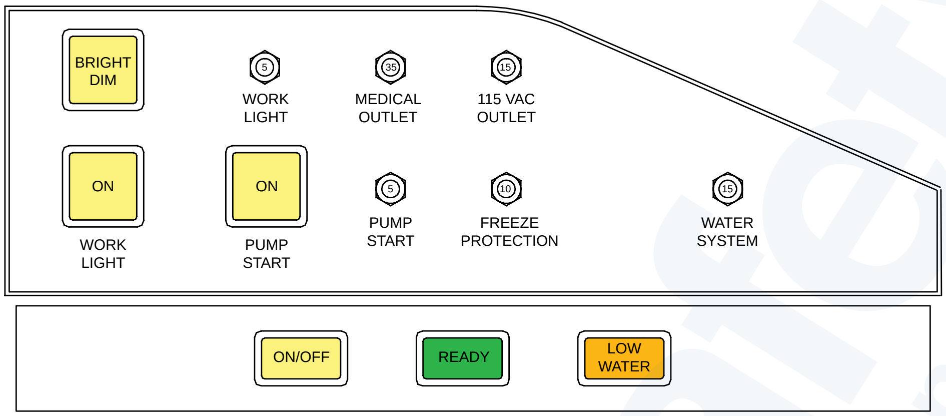
Two internal vents in the valve break the vacuum in the fill and overflow lines. This allows water trapped within the valve to drain away.

Boiler Assembly

The boiler assembly heats water from the tank for in-flight catering. It has a 6 liter tank, electrical box, a control panel, interconnecting plumbing, and electrical connections.

It uses 115 VAC 3-phase power to heat the water. The tank has the following items:

- Heating elements (6)
- Thermistors (2)
- Level sensor
- Float valve
- Pressure relief valve (65 psig)
- Thermal limiter.



WATER BOILER CONTROL PANEL

NOTE
Turn 90˚ to fill.
Pull and turn to drain.

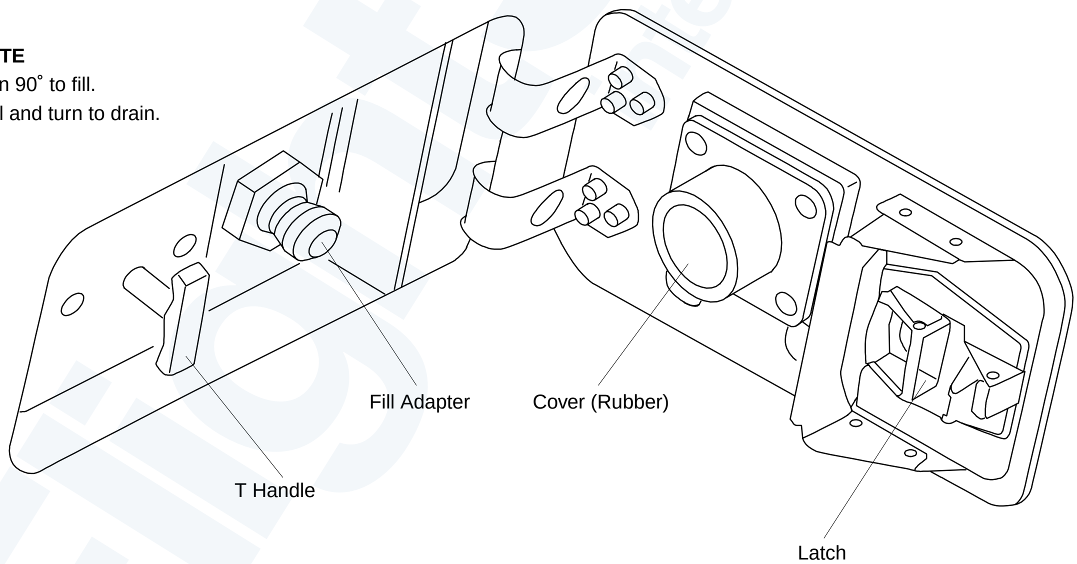


Figure 38-4. Service Panel

Ground Service Panel

Refer to Figure 38-4. Service Panel.

On aircraft with the pressure filling system (812CH00007), the ground service panel lets the ground service crew control the filling and draining of water.

The ground service panel has a fill adapter, a T-handle, and a cable control unit. The cable control unit has a control box installed behind the ground service panel with two cables coming out of it.

One cable is connected to the fill valve and the other is connected to the drain valve. Rotating the handle causes an inner cog in the control box to engage the cable and move it in or out. This action operates the fill valve and allows the system to be filled with water.

Pulling the handle out disengages the inner cog and allows the handle to engage an outer cog. Rotating the handle then operates the drain valve in a similar way.

CONTROLS AND INDICATIONS

Refer to Figure 38-4. Service Panel.

WORK LIGHT Switch

The work light is energized with the ON switch. The BRIGHT/ DIM switch controls the work light brightness.

PUMP START ON Switch (S3)

The pump (GL1-PM1) is energized with the ON switch.

A 5 amp circuit breaker protects the pump.

MEDICAL OUTLET

The medical outlet circuit is protected with a 35 amp circuit breaker.

FREEZE PROTECTION

The freeze protection and heater circuit is protected with a 10 amp circuit breaker.

WATER SYSTEM

The water system circuit is protected with a 15 amp circuit breaker.

Interfaces

The potable water system has the following electrical interfaces:

- The pump is powered from the R SECONDARY bus through a 5 A circuit breaker
- The start up 5 second time delay relay is powered from the R SECONDARY bus through a 5 A circuit breaker
- The boiler is powered from the right AC contactor box (three-phase 115-VAC)
- The freeze protection heaters are powered from the 115- VAC VARIABLE FREQUENCY RIGHT BUS through a 10 A circuit breaker.

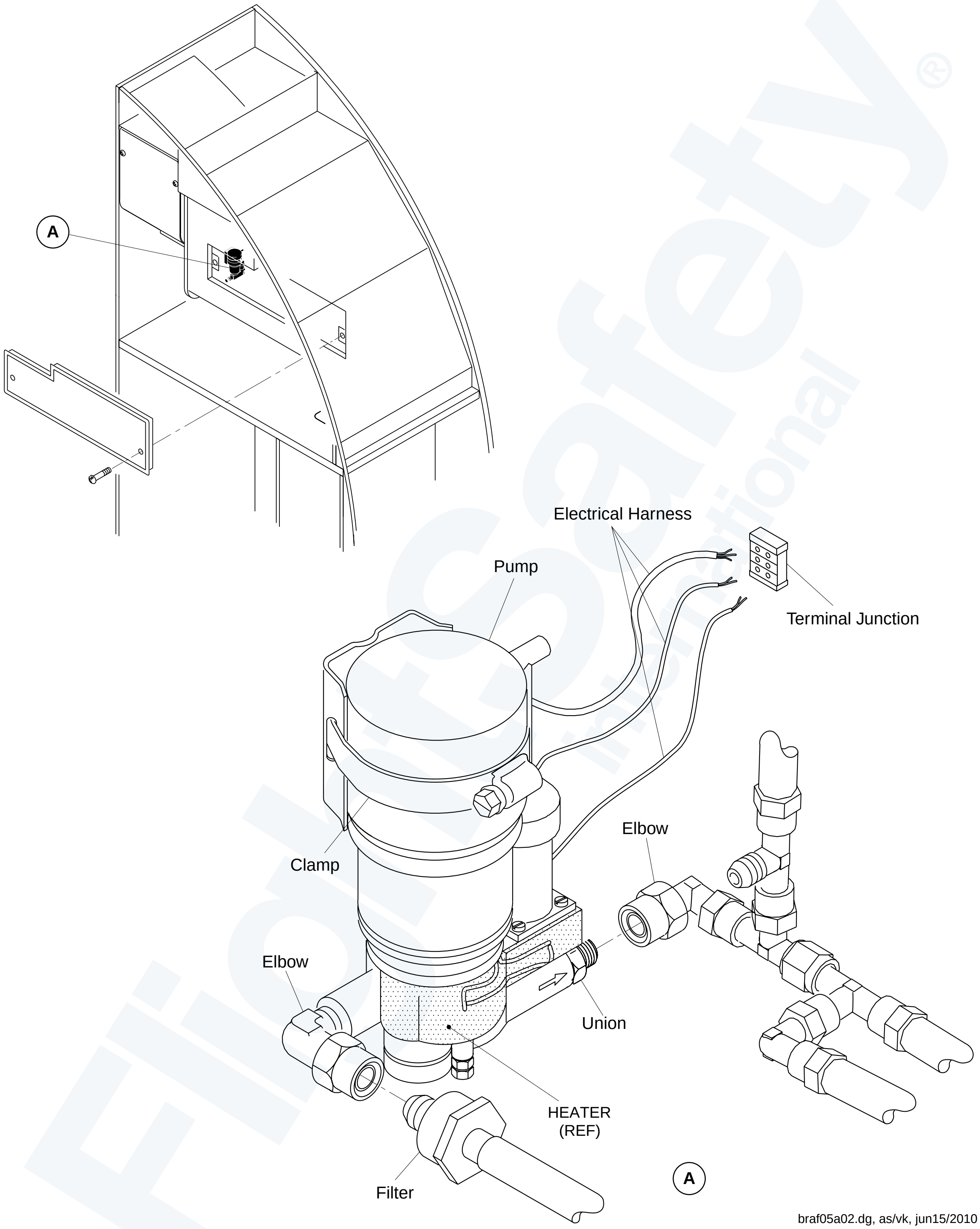


Figure 38-5. G1 Galley Pump

OPERATION

NOTES

Refer to Figure 38-2. Potable Water System Schematic.

A control panel on the face of the galley has an illuminated switch (ON/OFF), a LOW WATER light and a green READY light. Selection of the ON/OFF switch on the control panel in the galley area primes the electric motor driven gear pump.

The switch latches on a 5 second relay, which allows enough time for the pump to build up pressure and illuminate the amber ON light.

Power is supplied from the R SECONDARY bus. Once the pump is primed, the flow control will revert to a pressure switch in the pump. This pressure switch will only operate if the pump is first pressurized by priming.

Pump Operation

The PUMP START circuit breaker CB6 supplies the 28 VDC to the ON switch when the ON switch is pushed.

The PUMP START switch ON light illuminates.

The relay K1 causes the water pump to come on for 5 seconds.

When the tank runs empty, the LOW WATER advisory light illuminates.

The pump low level switch causes the pump to stop and de-energizes the boiler heater.

Water Pump

Refer to Figure 38-5. G1 Galley Pump.

The pump is installed in the G1 galley. While a heater and thermostat are bonded on the pump, they can only be replaced in overhaul shops.

Potable Water Boiler Control Box

Refer to Figure 38-6. Water Boiler Control Box.

The control box is installed inside the G1 galley. It controls the water heating in the water boiler.

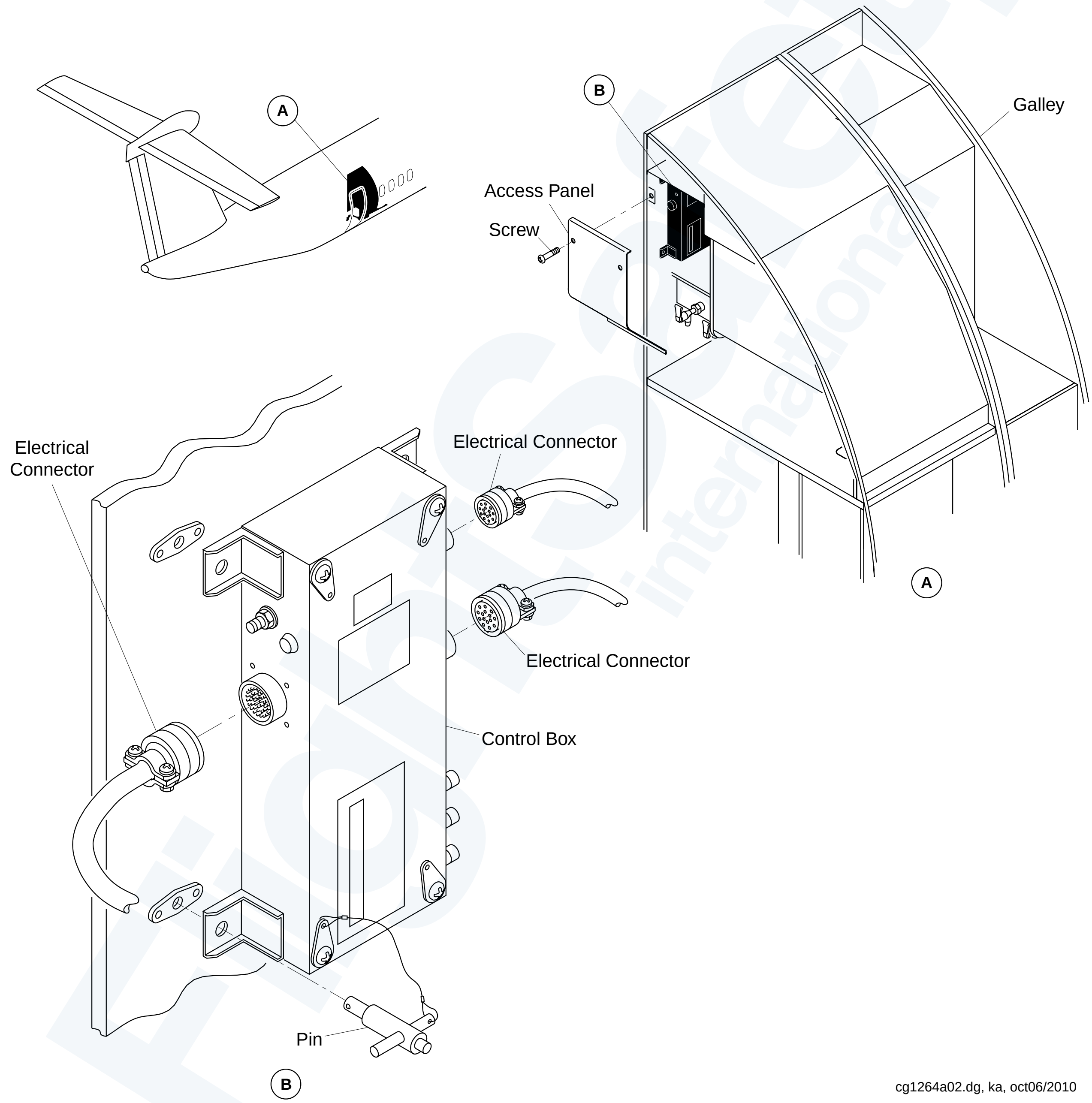


Figure 38-6. Water Boiler Control Box

Water Filter

Refer to Figure 38-7. Potable Water System Filter.

There is one filter in the potable water system. It is installed in the line between the water tank and the water pressure pump.

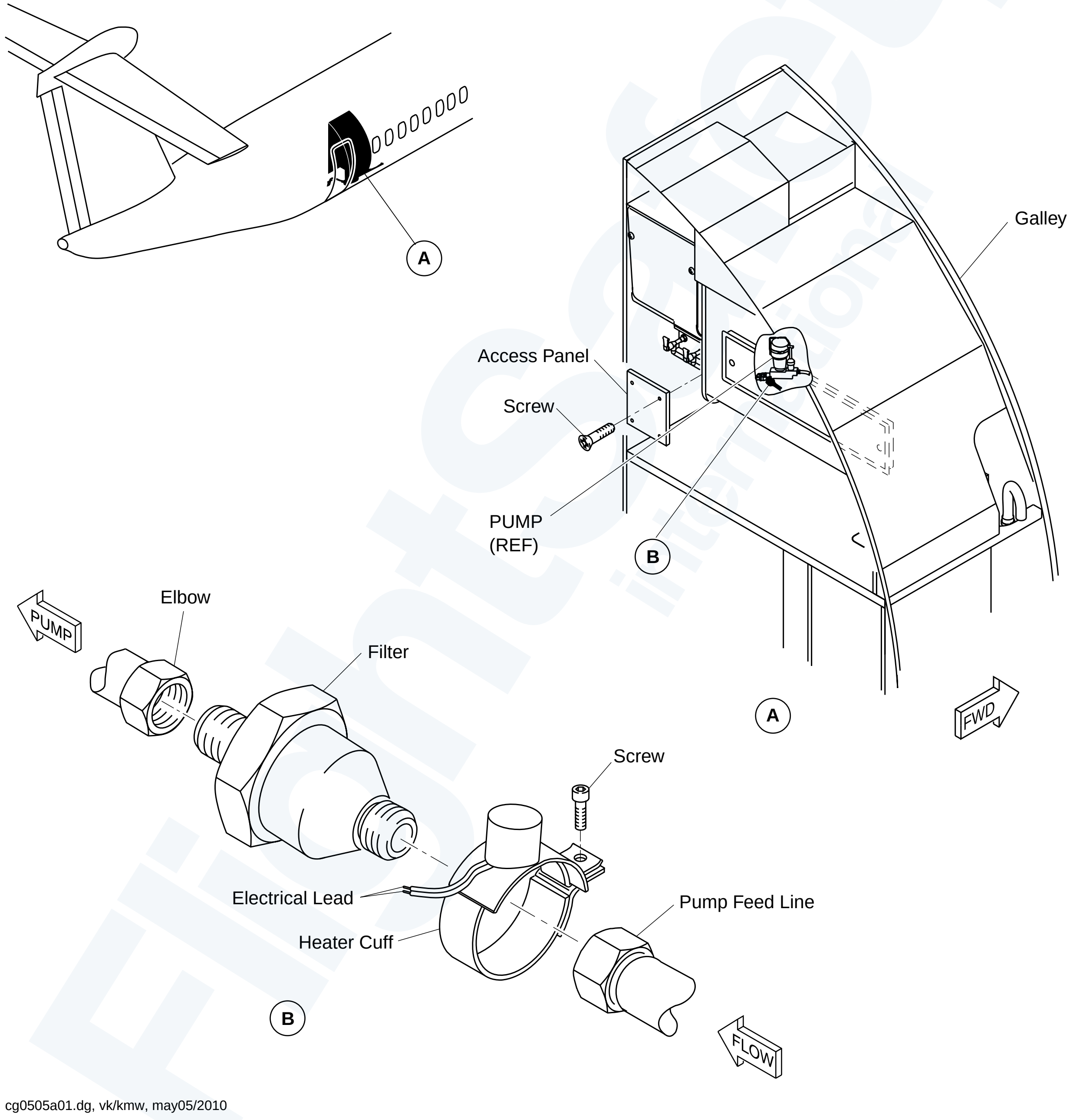


Figure 38-7. Potable Water System Filter

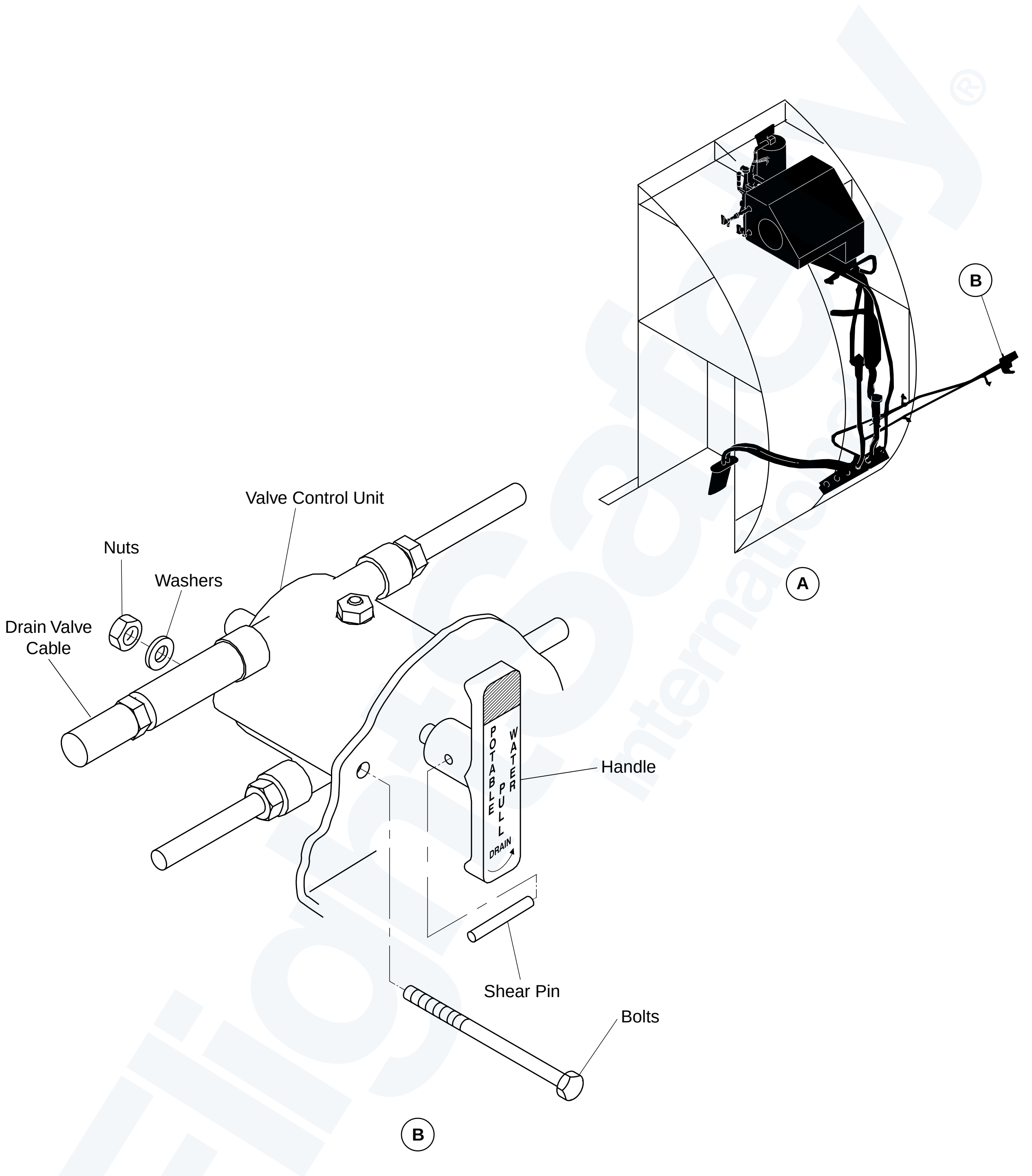


Figure 38-8. Potable Water System Control Valve Unit

Potable Water System Control Valve Unit

Refer to Figure 38-8. Potable Water System Control Valve Unit.

The valve control unit is installed below the floor and forward of the galley on the right side of the aircraft.

NOTES

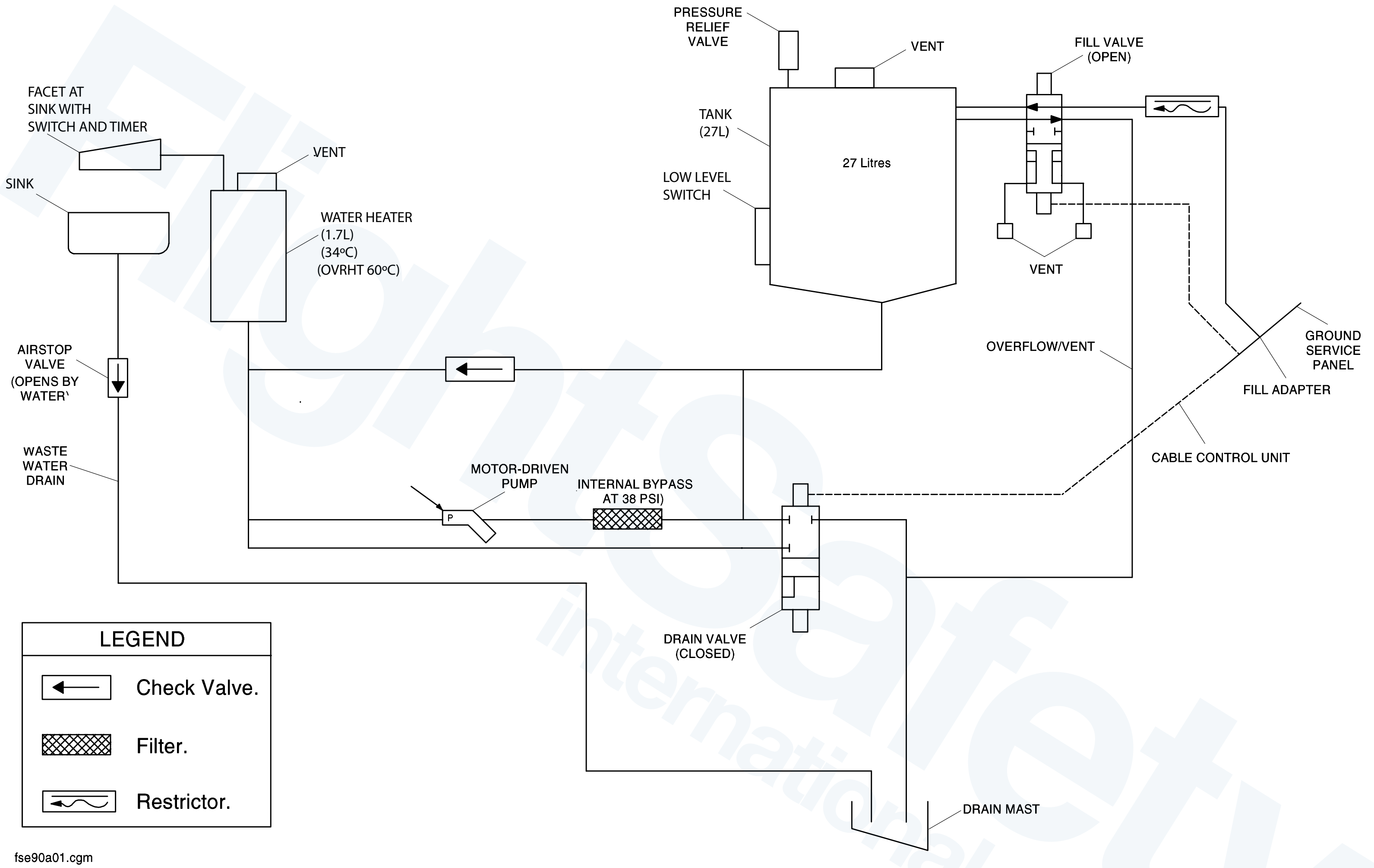


Figure 38-9. Lavatory Wash Water System Schematic

38-20-00 WASH WATER

INTRODUCTION

The wash water system supplies warm water to the lavatory sink for use by the passengers and crew.

GENERAL

The lavatory water system allows passengers to wash their hands in the lavatory with warm water. All components are installed below the sink in the lavatory compartment. Servicing is done through a service panel installed on the exterior fuselage.

The DC electrical system supplies power to the switches, valves and sensors which operate the lavatory water system. The AC electrical system supplies the power necessary to heat the system.

SYSTEM DESCRIPTION

The lavatory water system has a 27 liter water storage tank and a 1.7 liter heater tank. An electric motor driven gear pump supplies system pressure. The system has one faucet, with a built in switch, which activates the pump. The temperature of the hot water is approximately 34°C. All water lines and components are heated by electric heating elements.

Waste water is drained from the sink directly overboard through a drain mast installed in the underside of the aircraft. There is also a manual drain operated from the external service panel. This allows ground crews to drain the system for overnight storage in freezing conditions.

A ground service cart is used to fill the 27 liter storage tank. The supply line from the cart is attached to the fill adapter on the external service panel.

The T-handle on the panel is turned 90 degrees in a clockwise direction. The T-handle is connected to the valve control unit located just behind the service panel.

Once the valve is opened, water flows into the fill line for the tank. A flow restrictor in the fill line to the tank makes sure there is no increase in pressure in the tank during the filling process.

The tank has a pressure relief valve to prevent overpressure. A check valve in the fill line to the heater prevents any reverse flow from the heater to the storage tank. As the tank fills, the water opens the check valve allowing water to fill the heater vessel.

When the tank is approximately 90% full, water flows out the overflow line and drains through the drain mast. The overflow is visible to the servicing crew. It indicates that the tank is full.

Refer to Figure 38-9. Lavatory Wash Water System Schematic.

A switch and a control box are located near the faucet. Opening the faucet causes the switch to close which starts a time delay relay. The relay, which is located in the control box, energizes the pump for 5 seconds.

As the pump is energized, warm water flows out of the heater to the faucet. The pressure of the flowing water opens a float valve in the faucet allowing the water to flow into the sink. After 5 seconds, the pump is no longer energized and the water stops flowing.

Waste water from the sink is dumped overboard through the drain mast installed in the underside of the aircraft. An airstop valve in the sink drain line is opened by the force of the waste water.

When the water stops flowing, the airstop valve closes to prevent cabin pressure from venting through the drain during flight. A filter installed upstream of the pump prevents foreign objects from entering the pump and contaminating the system.

The heater uses a thermostat controlled electric blanket to maintain the water temperature at approximately 34°C. A thermal fuse limits the water temperature to a maximum of 60°C, in the event of a control thermostat failure.

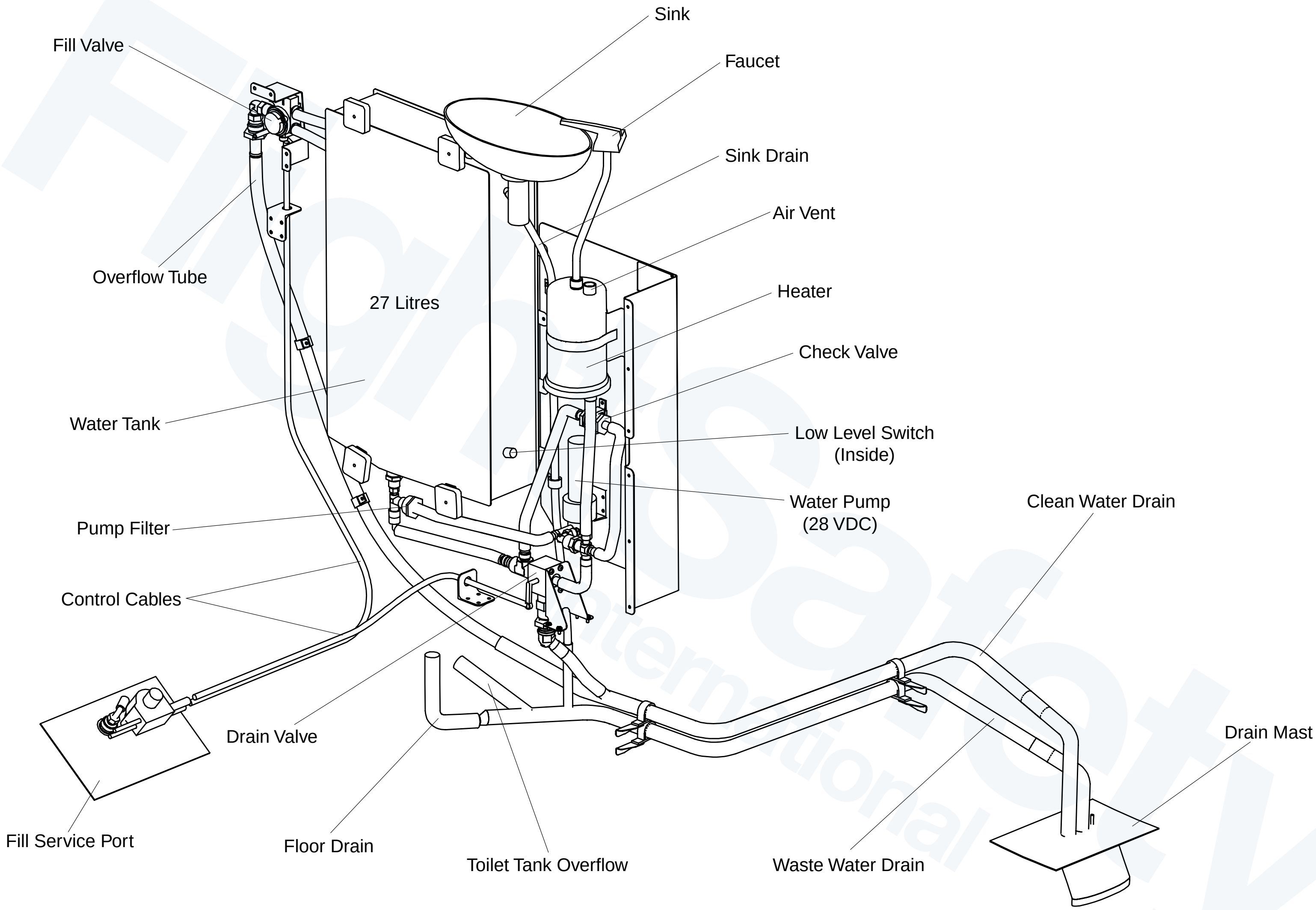


Figure 38-10. Lavatory Wash Water System Detail

A solid state low level switch shuts off the heater blanket and the pump when the water level is too low. The low level switch also turns on the LOW LEVEL POWER CUT-OFF light located on the lavatory control panel below the sink.

Heater cuffs protect the fill and drain valves, the pressure relief valve, the filter, the pump, the check valve and the tank from freezing during flight.

The cuffs are either molded directly to these components or are fastened by bolts. The drain mast and water lines have built in heaters. All the heaters have thermostats which control the temperature in the individual components.

The heater system is energized as soon as AC power is connected to the aircraft. This prepares the system for filling by preventing flash freezing of cold water coming into contact with cold components.

There is an option, for a freeze protection system to be run using external AC power. With this option the wash water and galley water systems do not have to be drained, if the aircraft sits for extended periods of time (overnight) in freezing conditions. The system is designed to work to a minimum temperature of -30°C. Below this temperature, the system must be drained.

COMPONENTS DESCRIPTION

Tank

Refer to Figure 38-10. Lavatory Wash Water System Detail.

The water tank provides storage for 27 liters of water, and provides the pump with positive head pressure.

The tank is installed above the floor, in a compartment under the lavatory sink. It is made from welded stainless steel. There is an access hole on the face of the tank to allow for cleaning by hand. A cover and gasket is attached with 8 fasteners, sealing the hole.

Fill Valve

Refer to Figure 38-10. Lavatory Wash Water System Detail.

In the open position, the fill valve allows water to flow from the fill line to the tank. It also allows overflow water from the tank to flow through to the drain mast. In the closed position, the valve isolates the tank from outside ambient pressure.

The heated fill valve is a 4-way, two-position valve. The valve is mechanically operated from the rotary control box located at the ground service panel.

Two internal vents in the valve break the vacuum in the fill and overflow lines, allowing water trapped within the valve to drain away.

Heating Element

Refer to Figure 38-10. Lavatory Wash Water System Detail.

The heating element converts electrical power to warm the valve to prevent freezing. A heating element with a built in thermostat is molded directly on to the valve body. The heater uses 115 VAC variable frequency and the temperature is controlled between 10°C and 30°C.

Check Valve

Refer to Figure 38-10. Lavatory Wash Water System Detail.

The check valve prevents the water in the heater from flowing back into the tank.

A check valve is installed in line between the tank and the heater. The head of water in the tank forces the water through the check valve into the heater during the filling process.

Drain Valve

Refer to Figure 38-10. Lavatory Wash Water System Detail.

The drain valve allows the entire system to be drained to prevent freezing in cold temperatures.

The drain valve is a two-position, 3-way ball valve, mechanically actuated from the rotary control box located at the ground service panel. A heating element is molded to the top of the valve. A built in thermostat controls the temperature of the heating element between 10°C and 30°C. The heater element uses 115 VAC variable frequency power.

Pump

The pump moves water from the tank to the lavatory sink.

The pump is a 28-VDC geared motor pump, with overtemperature protection. A magnetic clutch connects the motor to the gear drive. At a specified torque, the clutch disconnects and allows the motor to free spin.

The overtemperature protection is a self resetting thermostat located in the motor housing. The thermostat switch turns off the electrical power supply if the motor overheats. The pump is protected from freezing by an external heating cuff installed on the pump housing.

The pump has an internal bypass valve which prevents the system from overpressurizing during operation. The bypass valve allows water to recirculate from the valve outlet back to the inlet. The bypass pressure is set at 38 psi (262 kPa), and is adjustable.

Filter

The filter removes contamination from the water upstream from the pump.

The filter is sintered stainless steel woven wire mesh inside a stainless steel case. It has a nominal rating of 40 microns, to a 75 micron absolute. This prevents damage to the pump or clogging of other parts of the system. The filter is reusable and can be cleaned by backflushing or ultrasonic cleaning. The filter is heated by an external heater cuff.

Interface

The pump and control relay are powered from the 28 VDC R SECONDARY bus, through a 5 amp circuit breaker.

The heaters are powered from the 115 VAC VARIABLE FREQUENCY RIGHT BUS through a 10 A circuit breaker.

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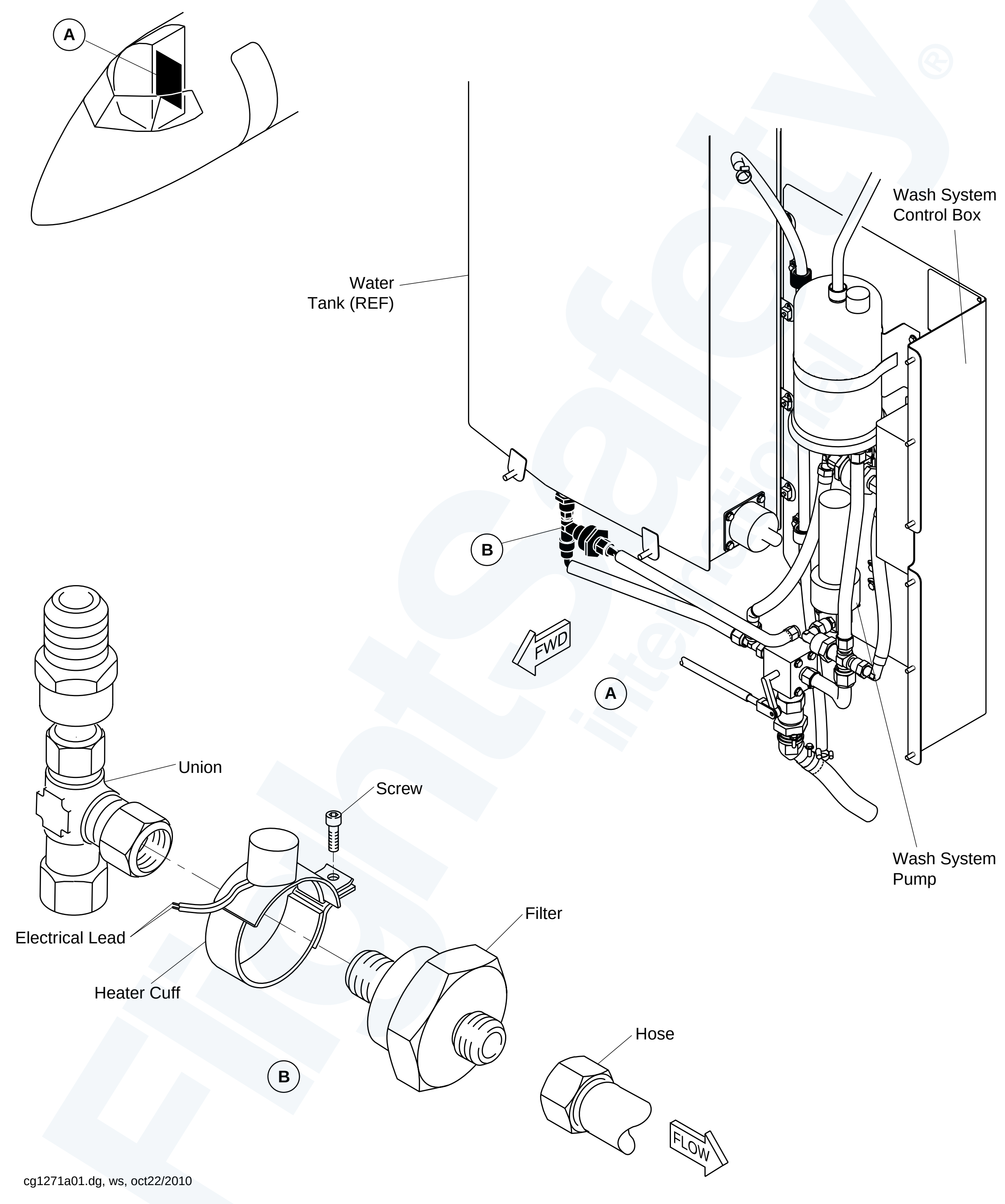


Figure 38-11. Lavatory Water Filter

OPERATION

Water Wash Operation

A switch and a control box are located near the faucet. Opening the faucet causes the switch to close which starts a time delay relay. The relay, which is located in the control box, energizes the pump for 5 seconds.

As the pump is energized, warm water flows out of the heater to the faucet.

The pressure of the flowing water opens a float valve in the faucet allowing the water to flow into the sink. After 5 seconds, the pump is no longer energized and the water stops flowing.

Wash Water System Servicing and Maintenance Practices

Refer to Figure 38-11. Lavatory Water Filter.

- Do not let the pressure increase to more than 40 psig (275 kPa) and the flow rate more than 20 gal/min (1.3 L/sec).
- It takes approximately 10 minutes to bring cold water to 34° + 4°/-6°C
- The yellow low level light on the lavatory control box comes on when a water tank low level condition is sensed

NOTE

The pump and the heater will not operate if a water tank low water condition is sensed and the low level light on the electrical control box is on.

- Follow AMM procedure to have the Wash Water System filter cleaned when it is clogged.

Crew Reports

Flight attendant complained that there is no water coming out from the washroom faucet. However the problem is still not rectified after the water pump is replaced and the wash water system serviced. What can be the probable cause? How would you isolate the root cause?

NOTES

38-30-00 WASTE WATER

INTRODUCTION

The lavatory waste water system supplies the water for the passenger washroom facilities on board the aircraft.

The system also stores the waste water after passenger use and allows it to be removed from the aircraft through a service panel.

GENERAL

Refer to Figure 38-12. Lavatory Detail.

The toilet unit is an electronically operated, recirculating flush toilet installed in the lavatory compartment. The lavatory compartment is manufactured in one piece. Any fluids spilled on to the floor are contained within the compartment and then drained overboard through the floor drain. The toilet unit is filled with clean water and is drained of waste water through an exterior service panel.

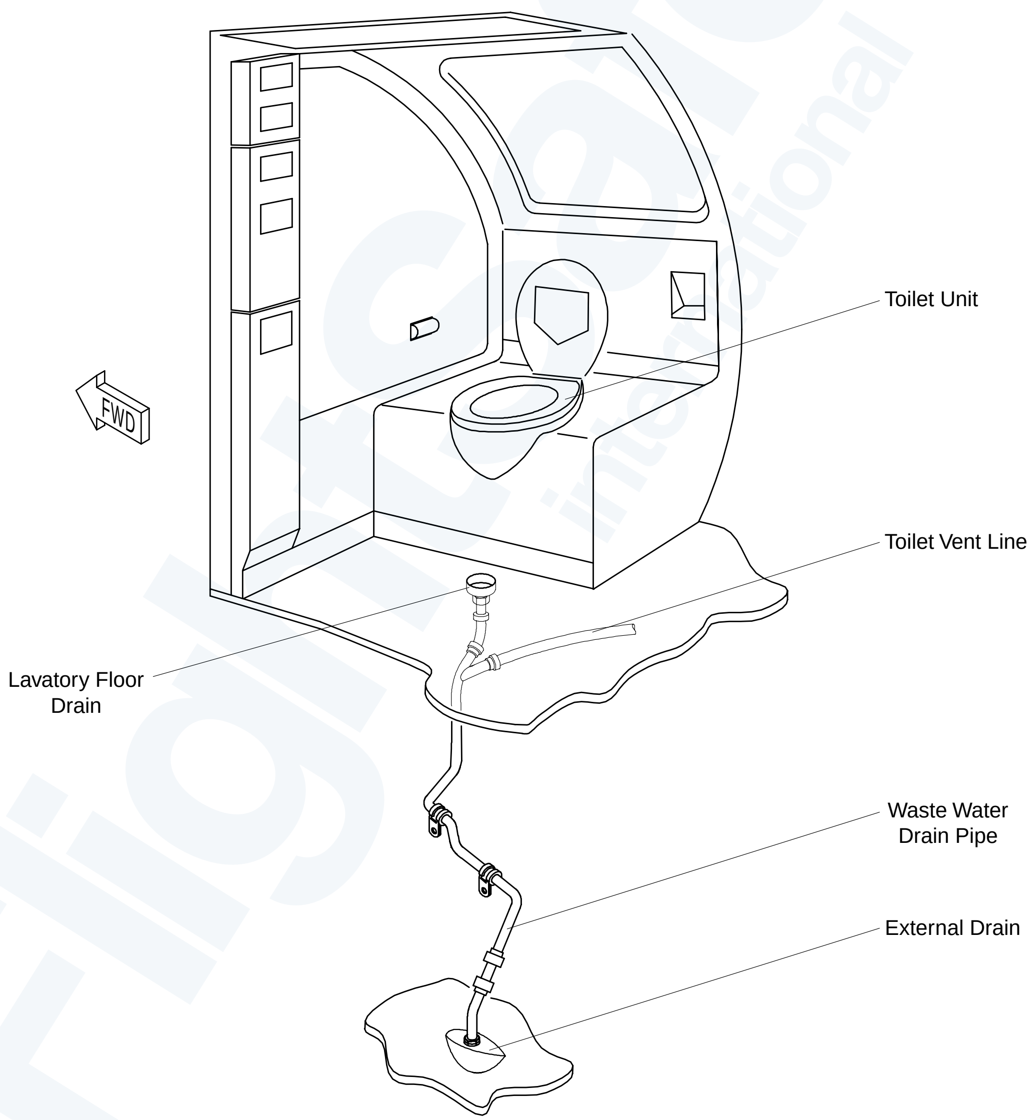


Figure 38-12. Lavatory Detail

SYSTEM DESCRIPTION

Refer to Figure 38-13. Lavatory Waste Disposal.

The toilet unit has a reservoir tank which holds the fluid used for flushing as well as the waste fluid. The unit also has a motor pump cartridge with a built in filter basket, a timer unit, toilet bowl, flush/fill line with a check valve, and a vent line with a muffler. The toilet unit is attached to the lavatory compartment floor by hold down rods and is covered around the base by a decorative shroud.

The toilet bowl is made from polished stainless steel and is installed on top of the fluid/waste reservoir. A flush channel allows a flow of fluid to cleanse the bowl during flushing. The unit is drained by dumping fluid and waste into a sewage drain which is directed to a service panel on the exterior of the aircraft.

Pushing the flush switch installed on the lavatory wall starts the flush cycle. When the flush switch is pushed, a timer unit, installed on top of the toilet unit, activates the 28 VDC motor driven pump for 5 seconds. The pump causes the flushing fluid to flow through the filter basket to the flush channel at the upper rim of the toilet bowl.

The flushing fluid flows down the inner side of the toilet bowl rinsing waste material directly into the reservoir.

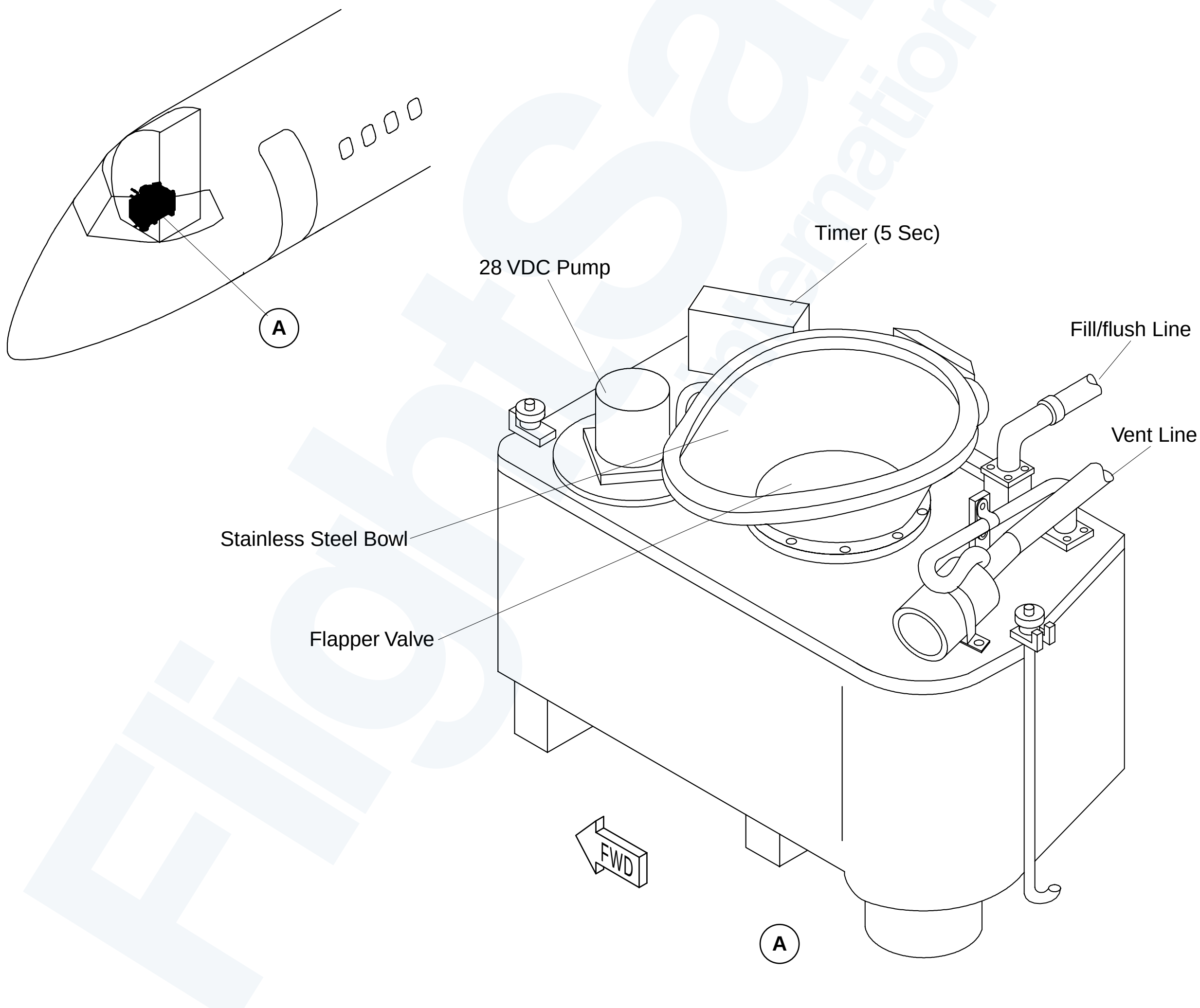


Figure 38-13. Lavatory Waste Disposal

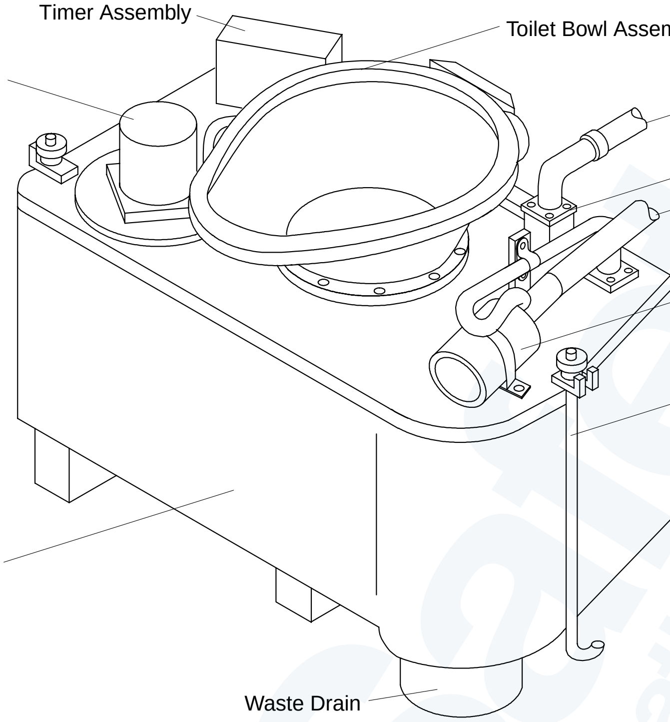
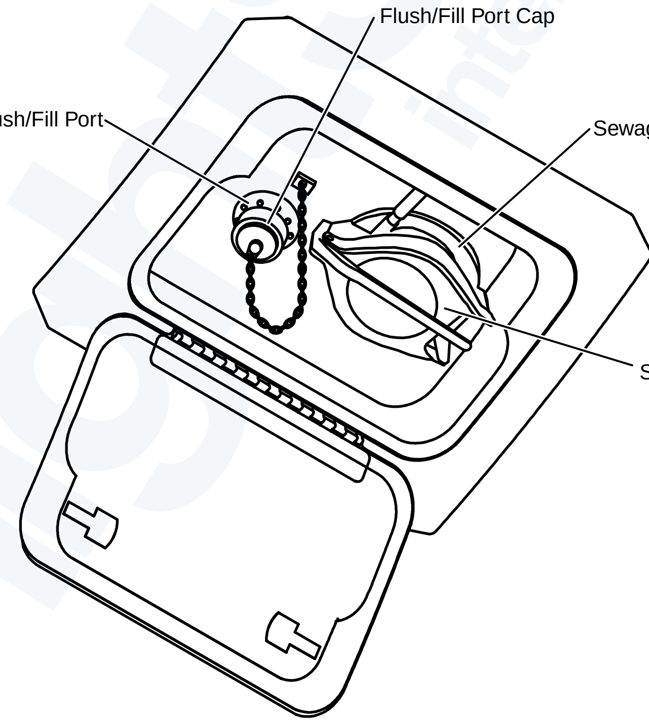


Figure 38-14. Lavatory Waste Disposal Detail



NOTE
Flush with 38 Litres.
Precharge with 9.5 Litres.

Figure 38-15. Lavatory Service Panel Detail

COMPONENTS DESCRIPTION

Refer to Figure 38-14. Lavatory Waste Disposal Detail.

Bowl Assembly

The shape of the bowl forces the water and waste to flow into the tank. The toilet bowl assembly is a polished stainless steel bowl installed on the top of the tank assembly. A one-way hinged flapper valve at the bottom of the bowl prevents waste and water from re-entering the bowl from the tank.

Tank Assembly

The top of the tank has provisions for the installation of the bowl assembly. The motor, pump filter assembly, the timer, the flush/fill check valve and the vent line elbow are all attached to the top of the tank. The tank and top assembly are made from fiberglass, with a built in structural reinforcement.

Pump and Filter

The pump and filter move filtered flush fluid from the tank to the bowl assembly. The motor pump and filter are an electric assembly installed in the top of the tank assembly. The filter is a TEF coated basket type filter that sits in the flush fluid. The motor has a built in thermostat which disconnects power when the temperature reaches a preset level. This protects against overheating of the motor if it runs continuously.

Timer Assembly

The timer assembly controls the flush cycle.

The timer assembly is an electronic unit that is installed on top of the tank assembly. When the toilet flush timer switch is pushed, the timer unit will operate the pump for 5 seconds.

Check Valve

The check valve allows water from the ground cart into the tank for cleaning and refilling. It does not allow water to flow back out through the fill tube. The flush/fill check valve is a mechanical valve installed in the top of the tank assembly.

Vent Line and Muffler

The vent line keeps toilet odors from entering the lavatory compartment while the muffler assembly, installed in the vent line, reduces the noise.

The tank is vented to the atmosphere through a vent line attached to the top of the tank assembly. The vent line is open to the atmosphere and makes a constant noise while the aircraft is in operation. The muffler assembly is installed in the vent line to reduce the noise.

CONTROLS AND INDICATION

Refer to Figure 38-15. Lavatory Service Panel Detail.

The lavatory waste water system is drained and filled through the lavatory service panel on the exterior of the aircraft. The sewage drain cover is opened and flush/fill cap removed. The coupling connector and flush/fill hoses from the ground cart are connected to the appropriate connectors. Push the inner valve latch to open inner valve flapper. The waste fluid and material drains into the ground service cart.

Once the tank has drained, the system is flushed with 10 gallons (37.85 liters) of pressurized water. The water enters the tank through the check valve to the pump inlet and a rotating spray nozzle. The pressurized fluid sprays through the nozzle and cleans the inside of the filter basket and tank sidewalls. Close the inner valve flapper and sewage drain cover. The inner valve flapper will close automatically when drain cover is closed and latched.

NOTE

Make sure the valve is clear of debris before closing inner valve flapper/drain cover.

The drain connector and flush/fill line are then disconnected and the sewage drain cover and flush/fill cap are replaced.

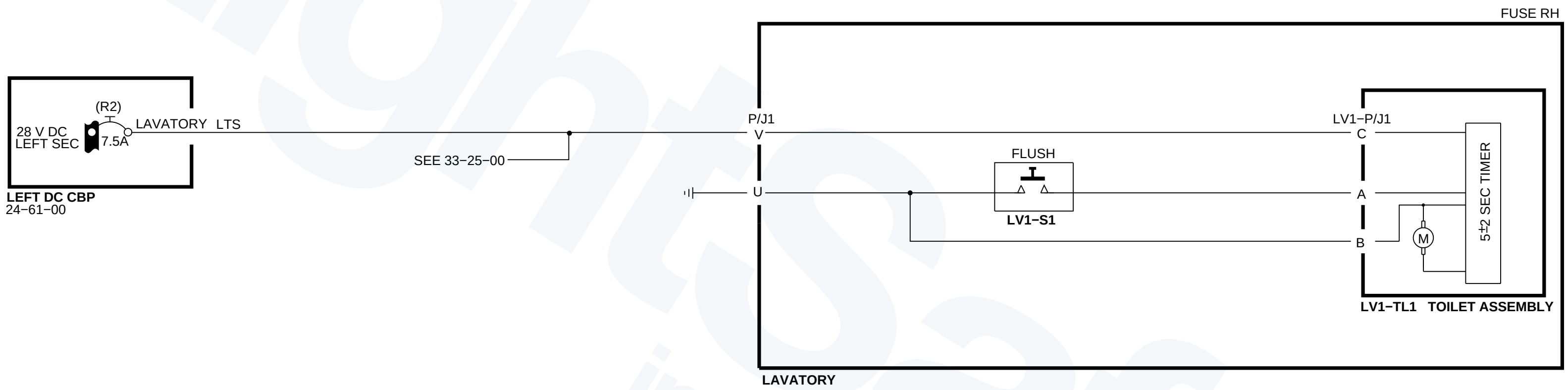


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Figure 38-16. Lavatory Waste Disposal System - Electrical Schematic

Interface

Refer to Figure 38-16. Lavatory Waste Disposal System - Electrical Schematic.

The toilet flush electric motor is powered from the L MAIN bus through a 5A circuit breaker. The electrical connector is attached to the timer unit.

Crew Reports

Flight attendant complained that the toilet does not flush. Which Line Replaceable Unit could have failed?

CDL FOR ATA 38

- The lavatory flush/fill port cap may be missing without performance penalty.

38-00-00 SPECIAL TOOL & TEST EQUIPMENT

- GSB1214001 Cart - Potable Water Service, Non-Winterized
- GSB1214002 Cart - Potable Water Service, Winterized, 110 VAC
- GSB1214003 Cart - Potable Water Service, Winterized, 220 VAC
- None specified Thermometer
- G601R121605-1 Gaseous Test Panel or equivalent
- 30682-12-12 Hose Barb Fitting
- 801-12 (or Kuri Tec K3150-12)
- GSB2400001 Multimeter - hand held

38-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 12-10-38-610-801: Servicing of the Wash Water System.
- AMM 12-10-38-670-802: Sanitization of the Galley Wash Water System.
- AMM 38-21-00-710-801: Operational Test of the Lavatory Water System.
- AMM 38-22-00-710-801: Operational Test of the Valve Control Unit Assembly.
- AMM 12-10-38-610-802: Servicing of the Waste Water System.
- AMM 38-32-00-710-801: Operational Test of the Drain Masts.